

Agentic Engineer Capstone -- Rubric Self-Check

Purpose

This self-check maps the current capstone evidence to the Canvas scoring criteria.

Statuses are intentionally conservative:

- **Complete** means repository evidence currently supports the criterion.
- **Partial** means meaningful evidence exists but additional final evidence is still needed.
- **Blocked** means completion currently depends on an external runtime dependency.
- **Pending** means the artifact or evidence has not yet been completed.

No criterion is marked complete based on estimated or invented results.

1. Workflow Scoping

Status: Complete

Evidence:

- docs/capstone-scope.md
- Stakeholder identified
- Trigger defined
- Inputs and outputs defined
- Acceptance criteria defined
- Failure modes identified
- Job-seeker/no-deployment path selected
- Existing Module 1-4 evidence and gaps inventoried

The selected workflow is an AI-assisted code review and governance pipeline with measurable review, quality, governance, and cost goals.

2. Sandboxed Environment

Status: Complete

Evidence:

- Dockerfile
- docker-entrypoint.sh
- scripts/run-agent.sh
- Role-specific read-only/read-write container behavior
- Memory mounted only where required
- Enforcement evidence in eval/enforcement-verification.md

The system runs in a containerized environment with explicit role-based workspace and memory boundaries. The README documents the capstone run path, required model credential, governed agents, preserved final-run evidence, and operational limitations for a new reviewer.

3. Quality Spec & Baseline

Status: Partial

Evidence:

- docs/prd.md
- eval/rubric.json

- historical Module 1 PRD/rubric/iteration evidence
- docs/calibration-log.md
- docs/capstone-impact-report.md
- docs/run-summary.md
- logs/capstone-final-human-approved-run.json

The rubric defines four judgment dimensions with pass thresholds. Measured baseline and comparison evidence exists for the deterministic conversion, including cycle time, model-token cost, review latency, structural checks, and repeatability. The successful final production-like run provides current execution evidence, including a 46.8-second full-pipeline runtime and recorded Planner, Reviewer, and Tester evaluation outcomes. The available historical Module 1 artifacts do not provide one complete baseline containing every capstone impact metric. The final transcript does not establish a trustworthy full-pipeline model-cost measurement, and one successful run is not sufficient evidence for a general defect-rate percentage. The impact report therefore keeps historical baseline evidence, deterministic-conversion measurements, and final production-like measurements separate rather than reconstructing or inventing missing values.

4. Agents, Skills & Memory

Status: Complete

Evidence:

- versioned role definitions under agents/
- persistent state in .memory/storage.db
- storage audit evidence
- reusable references under .memory/reference/
- role-scoped memory mounting
- docs/adr/ADR-004-memory-layout.md

The memory architecture separates persistent state, reusable references, task context, and role instructions.

5. Orchestration & MCP Tools

Status: Complete

Evidence:

- docs/orchestration-diagram.md
- docs/routing-and-tool-grant-map.md
- mcp-servers/storage/
- mcp-servers/retrieval/
- role-specific allow-lists
- classification ceilings
- retrieval ground-truth set
- retrieval quality report
- schemas/handoff.json

The Orchestrator delegates to scoped roles and MCP capabilities are not exposed globally.

6. Evaluation & Calibration

Status: Complete

Evidence:

- eval/test_deterministic.py
- eval/test_rubric_suite.py
- eval/rubric.json

- docs/holdout-task-set.md
- docs/calibration-log.md
- eval/run_holdout.py
- eval/run_regression.py
- regression and policy tests
- retrieval validation evidence
- logs/capstone-final-human-approved-run.json
- docs/run-summary.md

The project uses a two-layer evaluation design combining deterministic checks with judgment-based evaluation criteria. Calibration and regression evidence includes 7/7 deterministic checks passing before and after the handoff-validation conversion; zero output variance across three deterministic runs after conversion; 24/24 integrated regression checks passing; passing policy evidence; retrieval calibration and validation evidence; preserved holdout tasks; and a successful production-like model-backed workflow run. The final production-like run completed the Planner -> Implementer -> Reviewer -> Tester path and recorded successful evaluation outcomes. Measurement limitations, including the lack of a trustworthy full-pipeline model-cost value and the inability to generalize a defect-rate percentage from one successful run, are documented separately in docs/capstone-impact-report.md.

7. Governance, Security & CI/CD

Status: Complete

Evidence:

- docs/governance-policy.md
- docs/routing-and-tool-grant-map.md
- MCP allow-lists
- data-classification ceilings
- eval/test_policy.py
- eval/test_governed_files.py
- eval/red-team-prompts.md
- eval/red-team-results.md
- eval/enforcement-verification.md
- .github/workflows/ci.yml
- docs/ci-step-design.md

The tool-evolution drill demonstrated that permission drift is automatically detected by the policy suite. Known platform-level limitation: direct pushes from the authorized repository owner can bypass the configured pull-request and required-check rules. This is documented separately from repository-level enforcement.

8. Right-Tool Decisions & ADRs

Status: Complete

Evidence:

- docs/step-classification.md
- deterministic handoff validation
- deterministic routing decision history
- before/after latency and cost measurements
- ADR package under docs/adr/
- ADR coverage: rubric design; memory layout; MCP boundaries; subagent scoping and routing; governance policy; agent-to-deterministic conversion

Measured handoff conversion evidence shows 45 seconds to 0.2 seconds cycle time; \$0.003 to \$0 model-token cost; approximately 30 seconds to 5 seconds review latency; and 7/7 structural checks preserved.

9. Production Integration & Tool-Evolution Drill

Status: Complete

Evidence:

- successful production-like model-backed orchestration
- Planner -> Implementer -> Reviewer -> Tester execution
- deterministic lexical retrieval with classification-ceiling enforcement and persistent-storage activity
- Implementer storage write with downstream verification
- governance enforcement
- tool-evolution drill
- regression detection and recovery
- preserved transcript and audit evidence

The successful final human-approved production-like run completed in 46.8 seconds using 33,519 of the bounded 38,000-token workflow budget. Human approval was approve and completion_authorized was true. Tool-evolution results: Baseline 6 passed; Permission revoked 1 failed, 5 passed; Permission restored 6 passed. Governance enforcement was also demonstrated independently: an unauthorized documentation-writer write_entry request was denied by the storage MCP allow-list and recorded as authorization_denied. The earlier capstone-final-run-001 is retained as evidence of an integration failure that was diagnosed and corrected before the successful final run.

10. Iteration Narrative & Impact

Status: Complete with documented measurement limitations

Evidence:

- docs/iteration-log.md
- docs/calibration-log.md
- docs/capstone-impact-report.md
- docs/run-summary.md
- deterministic conversion before/after evidence
- successful production-like run
- tool-evolution drill

Measured deterministic-conversion impact includes cycle time 45 seconds -> 0.2 seconds; model-token cost \$0.003/run -> \$0/run for the converted step; review latency approximately 30 seconds -> approximately 5 seconds; structural quality 7/7 checks passing before and after conversion; output variance zero across three deterministic runs; and integrated regression 24/24 checks passing. The current final production-like pipeline completed in 46.8 seconds. In that successful run, 3 Planner review items, 4 Reviewer review items, and 1 Tester review item were approved, with no recorded rejected item. One successful run is not sufficient evidence for a general 0% defect-rate claim, and the transcript does not provide a trustworthy full-pipeline model-cost measurement. No unsupported values are estimated or invented.

11. Stakeholder Communication

Status: Partial

Evidence:

- docs/capstone-stakeholder-one-pager.md
- docs/capstone-architecture-writeup.md
- docs/capstone-impact-report.md
- business-value framing

- risk-reduction explanation
- successful production-like run evidence in docs/run-summary.md

The stakeholder one-pager and supporting reports describe the completed production-like workflow, measured deterministic-conversion impact, governance enforcement, business value, and remaining measurement limitations. Remaining requirement: the final five-to-ten-minute technical walkthrough must demonstrate the completed workflow, governance enforcement, evaluation evidence, right-tool decision, measured impact, and operational readiness.

12. Clarity & Flow

Status: Partial

The planned final narrative follows: 1. Problem and baseline; 2. Architecture; 3. Live workflow; 4. Governance; 5. Evaluation; 6. Right-tool decisions; 7. Impact; 8. Operations. Evidence documents follow this narrative structure. Final scoring depends on the completed walkthrough and presentation package.

13. Design

Status: Complete for document packaging; walkthrough remains

The required evidence PDFs have now been generated and updated, including the impact report, ops-ready runbook, and rubric self-check, with the ADR package and sanitization review also available. Architecture diagrams and tables should remain legible in the packaged artifacts. The remaining presentation item is the final five-to-ten-minute walkthrough/video and final submission review.

Current Readiness Summary

Criterion	Status
Workflow Scoping	Complete
Sandboxed Environment	Complete
Quality Spec & Baseline	Partial — historical baseline limitations documented
Agents, Skills & Memory	Complete
Orchestration & MCP Tools	Complete
Evaluation & Calibration	Complete
Governance, Security & CI/CD	Complete
Right-Tool Decisions & ADRs	Complete
Production Integration & Tool-Evolution Drill	Complete
Iteration Narrative & Impact	Complete with documented measurement limitations
Stakeholder Communication	Partial — final walkthrough remaining
Clarity & Flow	Partial — final walkthrough/package review remaining
Design	Complete for document packaging — walkthrough remains

Remaining Critical Path

1. Keep the final sanitization/secret review evidence with the submission package.
2. Keep the final test and repository-status evidence available for review.
3. Verify the required PDFs remain legible after upload.
4. Record/confirm the five-to-ten-minute technical walkthrough.
5. Perform the final Canvas submission review.

The production-like workflow, transcript and audit evidence, README run instructions, governance demonstration, tool-evolution drill, and required document packaging are complete.

Self-Check Conclusion

The repository contains substantial completed evidence for workflow design, sandboxed execution, specialized agents, memory architecture, orchestration, MCP boundaries, evaluation and calibration, governance, CI/CD controls, deterministic conversion, ADRs, production-like integration, and tool-evolution testing. A successful production-like run is preserved in `logs/capstone-final-human-approved-run.json` with supporting audit evidence and a final run summary.

Known limitations remain explicitly documented: the historical baseline does not contain every desired capstone metric, the final transcript does not establish a trustworthy full-pipeline model-cost measurement, one successful run is not sufficient to claim a general 0% defect rate, and the successful final run did not trigger the human-escalation path. These limitations are reported rather than filled with estimated or invented results.